A REMARK ON LOCAL LOGARITHMIC EMBEDDINGS FOR THE NEVANLINNA CLASS OF DIRICHLET SERIES

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ABSTRACT. Allu and Mondal asked whether the local integrals of $\log(1+|f|)$ over unit vertical segments in the right half-plane are bounded by a constant times the Nevanlinna gauge $\|f\|_0$, for Dirichlet series f in the Nevanlinna class $\mathscr{N}_u$. We show that the answer is no, already for exponentials of Dirichlet polynomials, which are bounded in the half-plane. The ratio of the two sides grows like $\sqrt{N}/\log N$ along the family $\exp(\sum_{n=2}^{N} n^{-s})$.

1. INTRODUCTION

Let $f(s) = \sum_{n\geq 1} a_n n^{-s}$ be a Dirichlet series whose abscissa of uniform convergence satisfies $\sigma_u(f) \leq 0$. Following Brevig and Perfekt [2] and Guo, Ni and Zhou [4], Allu and Mondal [1] consider the class $\mathscr{N}_u$ of such series with

$$\begin{aligned} \|f\|_0 &= \limsup_{\sigma\to 0^+} A(f,\sigma) < \infty, \\ A(f,\sigma) &= \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \log(1+|f(\sigma+it)|)\,\mathrm{d}t. \end{aligned} \tag{1}$$

Motivated by local Carleson embeddings for Hardy spaces of Dirichlet series, they posed the following question [1, Problem 4.13].

Problem 1 (Allu and Mondal)**.** *Is there a constant $C > 0$ such that every $f \in \mathscr{N}_u$ with $\sigma_u(f) \leq 0$ satisfies*

$$\sup_{\sigma>0}\sup_{\tau\in\mathbb{R}} \int_{\tau}^{\tau+1} \log(1+|f(\sigma+it)|)\,\mathrm{d}t \leq C\|f\|_0\,? \tag{2}$$

The purpose of the present note is to show that the answer is negative.

Theorem 1. *For $N \geq 3$ let $f_N(s) = \exp(\sum_{n=2}^{N} n^{-s})$. Then f_N is a Dirichlet series with $\sigma_u(f_N) = -\infty$, it is bounded in $\operatorname{Re} s \geq 0$, and*

$$\|f_N\|_0 \leq \log 2 + \frac{\sqrt{N-1}}{2\sqrt{2}}, \qquad \int_0^1 \log(1+|f_N(1/\log N + it)|)\,\mathrm{d}t \geq \frac{\pi(N-1)}{6e\log N}.$$

In particular, (2) fails for every C, also for the bounded Dirichlet series in $\mathscr{N}_u$.

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The mechanism is simple. On a short vertical segment near the real axis, all the terms n^{-s} with $n \le N$ point in nearly the same direction, so $\operatorname{Re}\sum n^{-s}$ is of order N on a segment of length of order $1/\log N$. The gauge, on the other hand, is a mean over the whole line, and by Kronecker's theorem it is the mean over a finite torus, where the terms are orthogonal and their sum has size $\sqrt{N}$.

2. Proof of Theorem 1

Put $g_N(s) = \sum_{n=2}^{N} n^{-s}$. Expanding the exponential and collecting products $n_1 \cdots n_k$, we find that $f_N = e^{g_N}$ is a Dirichlet series $\sum_m a_m m^{-s}$ with $a_m \ge 0$. For every real θ, $\sum_m a_m m^{-\theta} = \exp(g_N(\theta)) < \infty$, so the series converges absolutely and uniformly in $\operatorname{Re} s \ge \theta$. Hence $\sigma_u(f_N) = -\infty$, and $|f_N(s)| \le e^{N-1}$ for $\operatorname{Re} s \ge 0$. In particular $f_N \in \mathscr{N}_u$.

The gauge. Let $p_1, \dots, p_k$ be the primes not exceeding N, and for $z \in \mathbb{T}^k$ and a positive integer $n = \prod_j p_j^{\kappa_j}$ with all prime factors at most N write $\chi_n(z) = \prod_j z_j^{\kappa_j}$. Fix $\sigma > 0$ and put

$$X_\sigma(z) = \operatorname{Re} \sum_{n=2}^{N} n^{-\sigma} \chi_n(z).$$

Then $|f_N(\sigma + it)| = \exp(X_\sigma(p_1^{-it}, \dots, p_k^{-it}))$. Since $\log p_1, \dots, \log p_k$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem shows that $t \mapsto (p_1^{-it}, \dots, p_k^{-it})$ is equidistributed in $\mathbb{T}^k$ with respect to the Haar measure m. As $z \mapsto \log(1 + e^{X_\sigma(z)})$ is continuous,

$$A(f_N, \sigma) = \int_{\mathbb{T}^k} \log(1 + e^{X_\sigma}) \, \mathrm{d}m \le \log 2 + \mathbb{E}(X_\sigma)_+,$$

where $\mathbb{E}$ denotes integration against m and we used $\log(1 + e^x) \le \log 2 + \max(x, 0)$. By unique factorization the characters χ_n, $2 \le n \le N$, are distinct and nonconstant, and so are the products $\chi_n \chi_{n'} = \chi_{nn'}$. Writing $X_\sigma = (G + \overline{G})/2$ with $G = \sum_n n^{-\sigma} \chi_n$, we therefore have $\mathbb{E}G = \mathbb{E}G^2 = 0$ and $\mathbb{E}|G|^2 = \sum_n n^{-2\sigma}$, so

$$\mathbb{E}X_\sigma = 0, \qquad \mathbb{E}X_\sigma^2 = \frac{1}{2} \sum_{n=2}^{N} n^{-2\sigma} \le \frac{N-1}{2}.$$

Since $\mathbb{E}X_\sigma = 0$, the Cauchy–Schwarz inequality gives $\mathbb{E}(X_\sigma)_+ = \mathbb{E}|X_\sigma|/2 \le (\mathbb{E}X_\sigma^2)^{1/2}/2 \le \sqrt{N-1}/(2\sqrt{2})$. This holds for every $\sigma > 0$, and (1) gives the bound for $\|f_N\|_0$.

The local integral. Let $\sigma = 1/\log N$ and $0 \le t \le \pi/(3 \log N)$. For $2 \le n \le N$ we have $n^{-\sigma} \ge N^{-1/\log N} = e^{-1}$ and $0 \le t \log n \le \pi/3$, so

$$\operatorname{Re} n^{-\sigma - it} = n^{-\sigma} \cos(t \log n) \ge \frac{1}{2e}.$$

Hence $\log(1+|f_N(\sigma+it)|) \geq \operatorname{Re} g_N(\sigma+it) \geq (N-1)/(2e)$ on an interval of length $\pi/(3\log N) \leq 1$, and the integral over $[0,1]$ is at least $\pi(N-1)/(6e\log N)$.

Finally, the lower bound for the local integral divided by the upper bound for $\|f_N\|_0$ tends to infinity as $N \to \infty$. So no constant C satisfies (2) for all f_N. □

3. Concluding remarks

Weaker bounds. The same family shows that no bound of the form $\sup_{\sigma,\tau} \int_\tau^{\tau+1} \log(1+|f(\sigma+it)|)\,\mathrm{d}t \leq \psi(\|f\|_0)$ can hold with ψ nondecreasing and $\psi(x) = o(x^2/\log x)$ as $x \to \infty$. Replacing $\{2,\dots,N\}$ by a block $\{n; X \leq n \leq e^{1/8}X\}$ of integers and choosing the vertical segment where all the phases $t\log n$ are close to a multiple of 2π removes the logarithm, by the same argument: then the local integral is of order M, the number of terms, while the gauge is of order $\sqrt{M}$. Whether some bound $\psi(\|f\|_0)$ holds at all, for instance with $\psi(x)$ of order x^2, is a different question, which we do not address.

Relation to Hardy spaces. The local embedding problem for $\mathscr{H}^p$, which Chen, Fang, Guo, Hou, Shao and Zhou [3] have recently announced for every $p > 2$, concerns the line $\operatorname{Re} s = 1/2$ and the L^p norm. In (2) the segments approach the line $\operatorname{Re} s = 0$, where a Dirichlet polynomial with N terms can be of size N on a set of length $1/\log N$, while its mean square is only N; the logarithm does not compensate for this.

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References

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